

Assessing the resilience of stormwater ponds under climate change: A case study on Grace Lake, Florida, unveiling flood control and water quality implications

Rahman Davtalab^{a,*}, Saba Ghotbi^a, Raana Koushki^b, Caroline L. Reynolds^a

^a Singhofen and Associates Incorporated, 11723 Orpington St., Orlando, FL 32817, USA

^b School of Civil and Environmental Engineering, College of Engineering, Architecture and Technology, Oklahoma State University, Stillwater, OK, USA

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ABSTRACT

Stormwater ponds have emerged as a crucial solution in Florida to accommodate excess runoff from new developments and mitigate environmental pollution. Climate change affects rainfall and evapotranspiration, likely altering the hydrologic process. It is unclear how effective stormwater ponds will be in a changing climate, especially where surface water and groundwater interaction is highly expected. To address these concerns, this study employs a fully coupled surface water and groundwater model that was calibrated for the Grace Lake, located in central Florida. Utilizing the Multi-Model Ensemble method and GCM data, future rainfall projections were generated for the mid-century (2040–2070). The results indicate a projected 13 % increase in rainfall for the mid-century under the Representative Concentration Pathway 4.5 (RCP 4.5). This increase led to a rise of 13 cm in the seasonal high water elevation of Grace Lake, reducing its reliability as a flood control mechanism by 13 %. Moreover, downstream areas from the lake are expected to experience higher flood stages in the future, with the duration of road flooding extending from less than three days (base period) to approximately seven days (future period). This increase in surface runoff raises concerns for water quality, with nitrogen and phosphorus loads projected to surge up to 67 % in the future period.

1. Introduction

Hydrological models simulate the processes governing the movement and distribution of surface water and groundwater in a watershed from small scale to large scale (Noori et al., 2023). These hydrological models are fundamental for understanding and addressing issues related to stormwater management (Donnelly et al., 2022; Donnelly et al., 2024a), as well as ecosystem health and functions (Mahdian et al., 2023; Khosravi et al., 2023). Integrating surface water and groundwater movement provides a more comprehensive and accurate representation of the hydrological processes (Markstrom et al., 2008). Integrated hydrological models encompass a spectrum of modeling approaches, ranging from completely separated surface water (SW) and groundwater (GW) models to fully coupled SW-GW models (Fleckenstein et al., 2010). In completely separated models, SW and GW are modeled as entirely separate systems with their own set of equations and parameters (Winter, 1999), while in a partially integrated approach, SW and GW models are run independently and exchange information at specific time

steps, i.e., specified as loosely coupled models (Barthel & Banzhaf, 2016). These models provide some level of interaction between the systems but do not fully represent the complex mechanisms. On the other hand, fully coupled conceptual models simulate hydrological processes, from ground level to unsaturated topsoil and deeper saturated soil layers, using physical laws (Guevara-Ochoa et al., 2020). These tools provide fully Integrated SW and GW modeling to explore spatially distributed SW and GW interaction for wetlands, depressions, and other land features within small areas to large-scale watersheds (Jaros et al., 2019).

The separated and loosely coupled models are often computationally efficient and sufficient for simple systems with minimal surface and groundwater interactions (Furman, 2008). While these models may not be appropriate for complex systems, particularly when an accurate representation of interactions is essential, the fully Integrated SW/GW models (ISGM) can capture the complex interactions between SW and GW systems which leads to an accurate representation of real-world hydrological processes. SWAT-MODFLOW (Bailey et al., 2016; Liu

* Corresponding author.

E-mail addresses: Rdavtalab@saiengineers.com (R. Davtalab), saba.ghotbi@saiengineers.com (S. Ghotbi), Rana.koushki@gmail.com (R. Koushki).

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et al., 2020) and GSFLOW (Markstrom et al., 2008) can be specified as loosely coupled models, and Interconnected Channel and Pond Routing (ICPR) (Streamline, 2022) and HydroGeoSphere (Brunner & Simmons, 2012) are specified as fully coupled models.

It is worth mentioning that the ISGMs face two major constraints: the lack of spatially and temporally appropriate data for model parametrization and validation (Barthel & Banzhaf, 2016); and the significant execution time, especially for long-term continuous simulations (Goderniaux et al., 2009). However, over the last couple of decades, numerous organizations including USGS, NASA, FDEP, and others, have actively observed and recorded data from the Earth and atmosphere. This effort has substantially increased the spatial and temporal coverage of data available for modeling purposes. Additionally, advancement in remote sensing technologies such as Lidar sensors, have improved the accuracy of geographic information, including digital elevation models. On the other hand, the emergence of new computers equipped with high-speed CPUs and advanced graphics has addressed some of the challenges associated with time-consuming mathematical computations. These advanced computing systems significantly accelerate processes in continuous simulation and sensitivity analysis, leading to faster and more efficient calibration and validation results in modeling (Kurtz et al., 2017; Miller et al., 2018; Fattahi Marnani & Cuocci, 2022; Cuocci et al., 2023).

Climate change has the potential to alter both the volume of rainfall and its spatiotemporal distribution, exerting significant impact on the water cycle and hydrological process (Clifton et al., 2018; Dagbegnon et al., 2016; Donnelly et al., 2024b; Guevara-Ochoa et al., 2020; Noori et al., 2023; Persaud et al., 2020; Koushki et al., 2023a, 2023b). Surface water interacts with groundwater through wetlands, streams, and significant depressions; therefore, climate change effects on surface water are expected to affect groundwater indirectly (Dragoni & Sukhija, 2008; Amanambu et al., 2020). In order to explore the impact of climate change on surface water and groundwater, the General Circulation Models (GCMs) have been widely used to project meteorological parameters such as temperature, rainfall, and evapotranspiration (Dibike & Coulibaly, 2005; Charles et al., 2007; Ouyang et al., 2015; Pour et al., 2018). The projected climate parameters are used as input data for the hydrological model to project future water cycles (Wiley et al., 2010). GCMs are coarse resolution models which are required to be downscaled for local studies (Davtalab et al., 2014; Hassan et al., 2015; Wang et al., 2015). Several statistical methods have been developed to downscale GCMs over the U.S. continent (Wood et al., 2004; Maurer et al., 2010; Abatzoglou & Brown, 2012). Downscaling techniques rely on statistical methods that establish empirical connections between GCMs at large scales and local climate variables at fine scales. In recent years, there has been a notable incorporation of machine learning into climate modeling, specifically in the realm of statistical downscaling. This integration has marked substantial advancements in comprehending intricate relationships and non-linear patterns within climate variables. However, maintaining the model accuracy and reliability by machine learning-based methods are constrained by existing substantial historical data for effective training (Yeganeh-Bakhtiary, et al., 2022). After years of dedicated studies on climate change, numerous GCMs have been developed and extensively utilized. For instance, the Coupled Model Intercomparison Project Phase 5 (CMP5) has 50 GCMs with a wide range of projections for precipitation and temperature (Hayhoe et al., 2017). Therefore, climate change studies face significant uncertainty caused by different GCMs. Several studies have highlighted that the application of a sub-set of GCMs or Multi-Model Ensemble (MME) has the potential to address the uncertainties associated with GCMs in projecting future climate (Gohari et al., 2013; Sa'adi et al., 2017; Salman et al., 2018; Pour et al., 2018; Khan et al., 2018; Wang et al., 2021; Shetty et al., 2023). MME also improves the reliability of a future projection since it is established based on the results of multiple models instead of a single model (Ahmed et al., 2019).

The concurrent utilization of fully coupled ISGM and climate change

scenarios, which are corrected at regional and local scales, will improve the robustness of projects and facilitate decision-making in complex hydroclimatic systems (Sophocleous, 2002; Fleckenstein et al., 2010). Due to the extensive data requirements of ISGM and the considerable uncertainty within the present and future hydroclimatic system, there is a limited understanding of the interaction between SW and GW under climate change scenarios (Thompson et al., 2015; Pholkern et al., 2018; Jaros et al., 2019; Persaud et al., 2020; Guevara-Ochoa et al., 2020).

In the United States, Florida holds the third-highest position in terms of the number of claims filed by National Flood Insurance Program (NFIP) over the past four decades (NFIP, 2015). Stormwater ponds are a sustainable solution to stormwater treatment and flood control in Florida (Joyce et al., 2017; Davtalab et al., 2020; Davtalab & Ghotbi, 2022) and other similar climates (Manzo et al., 2020). Despite the similarity with natural ponds, stormwater ponds receive more frequent and higher amounts of stormwater due to the broad impervious area for residential and industrial land categories and consequently receive higher loads of contaminants like nitrogen and phosphorus (Hassall, 2014). Stormwater ponds are typically designed considering static hydroclimatic conditions, and the impact of climate change on their performance remained unclear. To gain precise insights into the performance of stormwater ponds in environments like Florida, where there are high levels of surface water and groundwater interactions, it is essential to utilize coupled ISGM. The primary goal of the Grace Lake study was to evaluate the stormwater system reliability for Grace Lake and the adjacent area regarding flood control and water quality. The coupled ISGM was applied to simulate surface water and groundwater interaction under current and future climates. The multi-model ensemble method was applied to address the uncertainty of GCMs for future projection.

This study, which integrates ISGMs and multi-model ensembles, represents a significant step forward in addressing these pressing issues. By evaluating stormwater system reliability in the face of climate change, it pioneers a comprehensive approach that considers both water quantity and quality. As such, this study fills a critical gap in our understanding of complex hydroclimatic conditions and equips decision-makers with valuable insights for sustainable water management practices.

2. Material and method

2.1. Study area

Florida has a semi-tropical climate and receives substantial rain from tropical storms and hurricanes during the wet season (May through Oct.). The project area is a mix of developed and undeveloped (rural) areas. The drainage system includes a series of interconnected channels, ponds, and wetlands that act as a stormwater management system. The primary flow path initiates from the west and southwest and continues towards the east. After collecting stormwater from an area of 50 km², the system ultimately discharges into Lake Jesup. The total modeled area for this project was about 50 km² with a narrowed focus on Grace Lake and a few neighboring wetlands in the reduced study area. A sinkhole was formed in Grace Lake along the southern perimeter around the year 1986. This sinkhole served as a considerable sink for the lake until around 2015, at which point it collapsed and partially closed. This event has resulted in a diminished capacity for natural storage recovery in the lake. Over the past few years, the water level in the lake has gradually risen and remained relatively high. The elevated water levels have resulted in tree mortality at various sites and nuisance flooding, exemplified by road inundation downstream of Grace Lake following Tropical Storm Irma in September 2017.

2.2. Study framework and steps

This study aimed to evaluate the interaction of surface water and groundwater in a very dynamic and complicated hydroclimatic

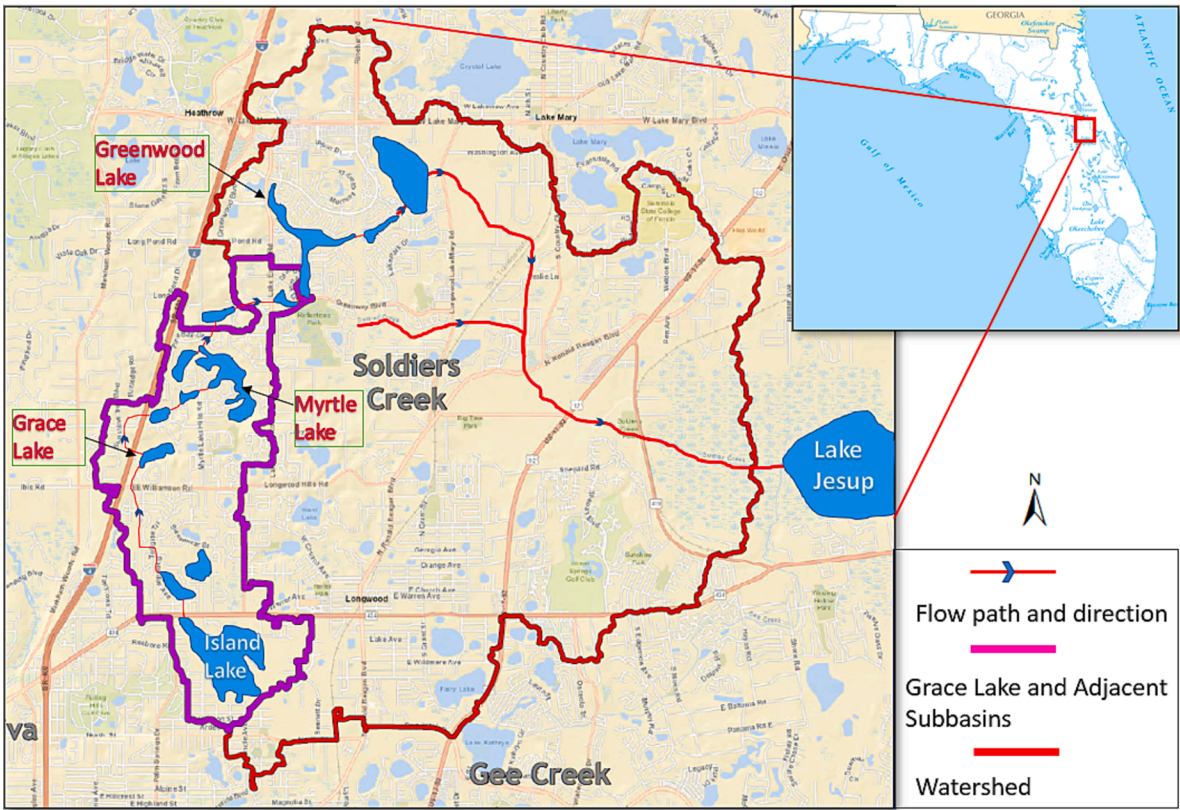


Fig. 1. Location of Grace Lake, related watershed, and drainage major system.

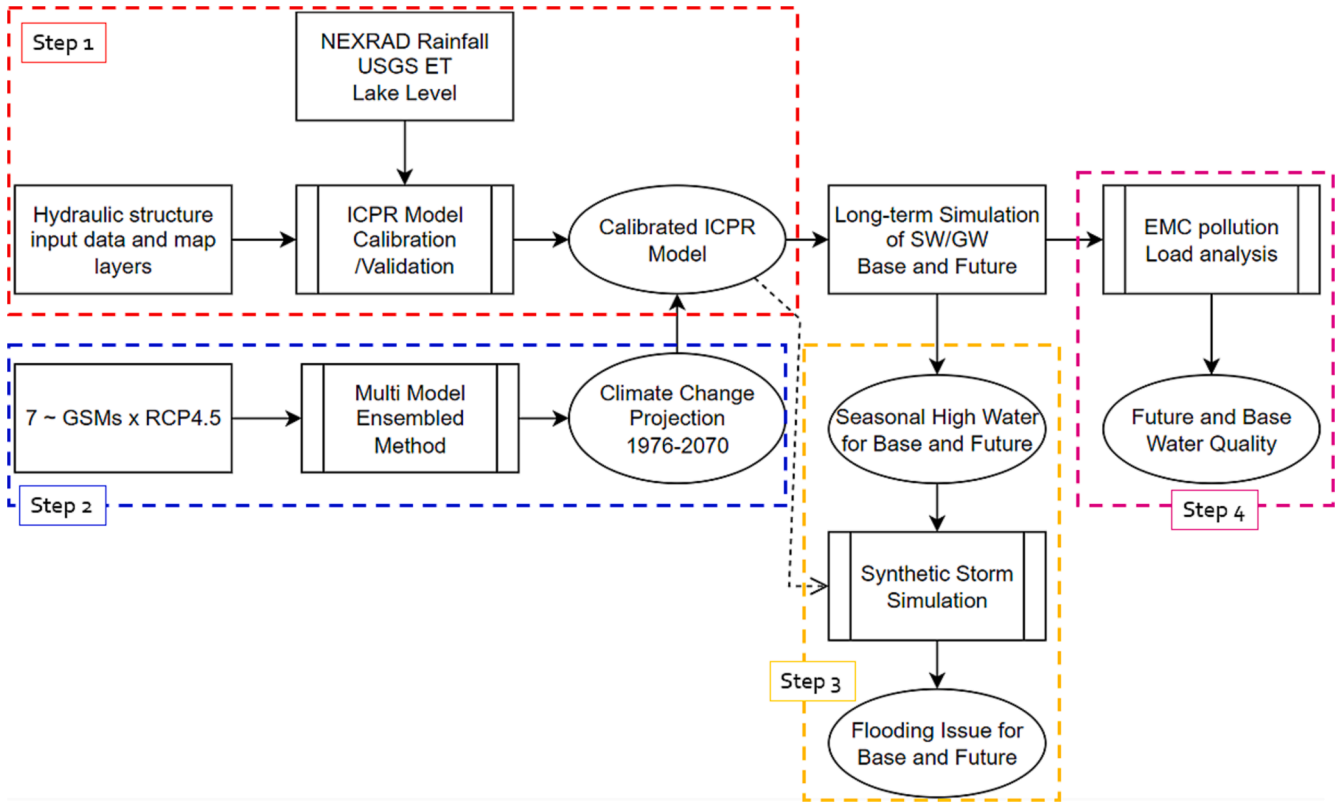


Fig. 2. Major steps and methodology.

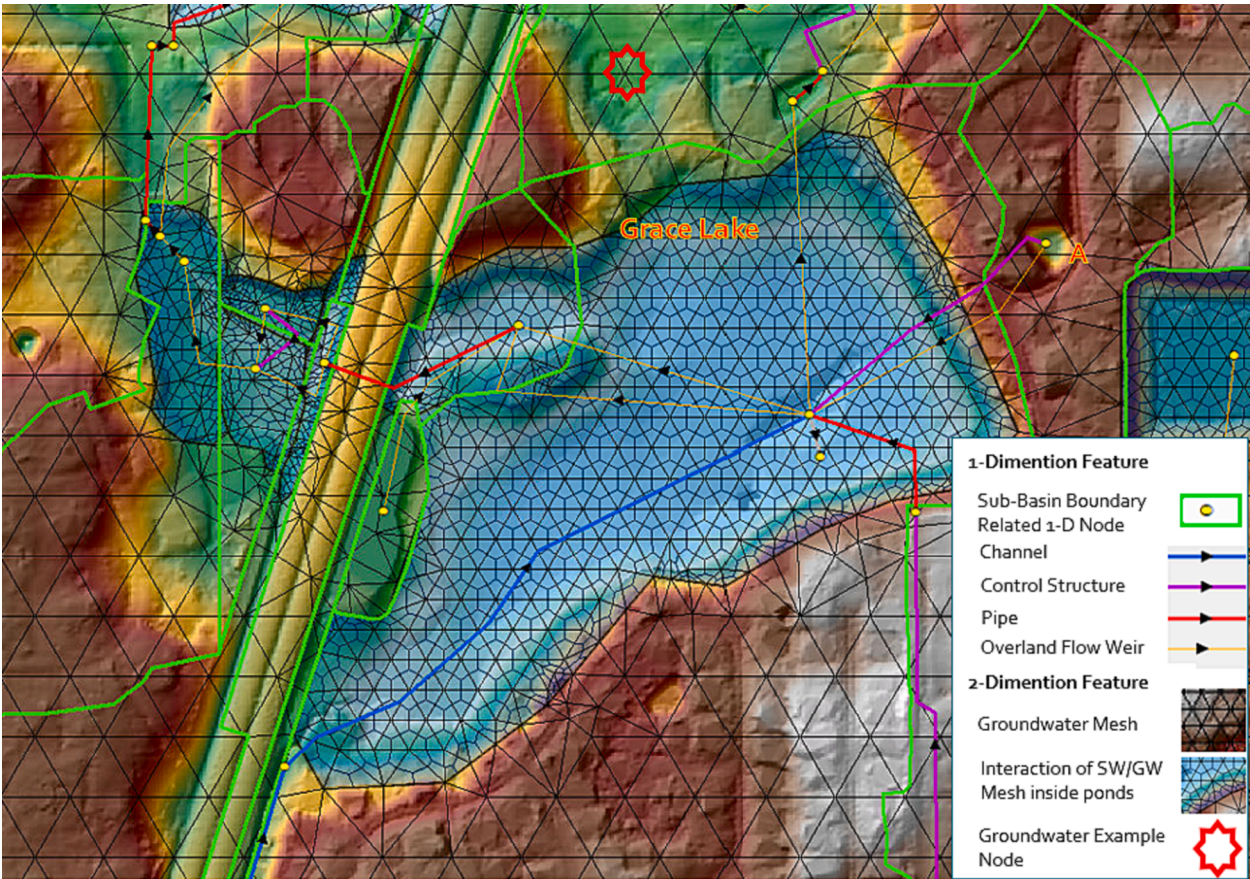


Fig. 3. Excerpt of surface water/groundwater model network in the vicinity of Grace Lake.



Fig. 4. Road flooding at Myrtle Lake Hills Road, downstream of Grace Lake (By Seminole County; IRMA, 2017).

environment. The evaluation covered both water quantity and quality for short-term and long-term periods. The study had four distinct steps as shown in Fig. 2. In step 1, an integrated surface water/groundwater model was developed using ICPR software. The model includes 1-Dimensional and 2-Dimensional approaches, and the model parameters were calibrated and validated. In step 2, projected rainfall and temperature data were downloaded from seven different GCMs, and the uncertainty of GCM data was addressed using an ensembled method. The daily rainfall and temperature data were generated for the time horizon 2060 (2040–2070). The observed data for the base condition (1976–2005) and future data were fed into the calibrated model from step 1, and continuous simulations were run for the base and future conditions. The Seasonal High-Water Level (SHWL) for the base and future conditions resulted from the long-term simulation. The estimated SHWL for the base and future conditions and current and future rainfall intensities were applied to simulate synthetic storms, and the flood risk analysis was conducted for both the base and future conditions in step 3. Finally, the surface water pollutant transfer was estimated using the results of long-term simulation and Event Mean Concentration (EMC) for both the base and future conditions in step 4.

2.3. Integrated surface water groundwater model

As part of this study, a fully integrated surface water groundwater model was developed to evaluate the effect of climate change on water resources and the reliability of stormwater systems in terms of quality and quantity. The Interconnected Channel and Pond Routing (ICPR version 4.07.08) model was used in this study (Streamline, 2022). ICPR simulates excess rainfall and infiltration to the soil column by modeling the vertical water movement through the unsaturated soil column and the horizontal water movement through the saturated soil zone. The surface water component routes runoff through a wide range of stormwater facilities, including channels, culverts, ponds, and pump stations. While groundwater uses a 2-Dimensional (2-D) approach, the surface water overland flow can be modeled using either the 1-Dimensional (1-D) or 2-D approach, or a combination of both. In this study, a combination of 1-D and 2-D approaches was employed; 2-D modeling was used for large ponds and significant depression areas, and the 1-D approach was used for the rest of the watershed. This method facilitates surface water and groundwater interaction in areas where significant GW and SW interaction occurs. Utilizing a partial 2-D modeling approach improves simulation results without significantly increasing simulation time, as it avoids inclusion of unnecessary areas in the 2-D approach. ICPR has several methods to simulate excess rainfall. The Green-Ampt method was utilized in this study since the method has the potential to perform both single-event and continuous simulations (Green & Ampt, 1911). The Green-Ampt method tracks water movement within the soil during wet and dry periods (Davtalab et al., 2020; Streamline, 2022). Fig. 3 provides an excerpt of the ICPR model network in the vicinity of Grace Lake, illustrating the combination of 2-D and 1-D approach modeling. This approach captures the interaction between surface water and groundwater in this area.

2.4. ICPR model calibration and validation

Green-Ampt parameters for the Grace Lake project were calibrated for 2017. In this year, Hurricane Irma occurred from September 9 through 11 and dropped approximately 25.4 cm (10 in.) of rainfall, causing road flooding downstream of Grace Lake. Fig. 4 shows road flooding at Myrtle Lake Hills Road, located downstream of Grace Lake. It should be noted that the Grace Lake sinkhole was resealed in 2015, which caused the water level at the beginning of the storm to be several feet higher than before 2015 and thus intensified the effect of flooding caused by Hurricane Irma. The Green-Ampt parameters were validated for 2018 as a normal year. However, the calibration and validation periods are short because of the shortage of observed data after 2015. The

Table 1

Metrics for evaluation of parameter calibration and validation.

Statistic	Range	Optimal Value	Calibration	Validation
Coefficient Determination (R^2)	0 to 1.0	1.0	0.94	0.92
Nash-Sutcliffe Efficiency (NSE)	$-\infty$ to 1.0	1.0	0.76	0.87
Mean Absolute Error (MAE)	0 to ∞	0.0	15.7 cm	6.2 cm
Root Mean Square Error (RMSE)	0 to ∞	0.0	18.5 cm	8.7 cm
Max. Flood Stage Difference (MFS D)	$-\infty$ to ∞	0	12.2 cm	6.7 cm

15 min NEXRAD rainfall data was ordered from ST. Johnes River Water Management District (SJRWMD) and used for model calibration and validation. The daily Potential Evapotranspiration was downloaded from USGS website: <https://www.usgs.gov/centers/cfwsc/science>.

Calibration and validation of the model were primarily carried out for Grace Lake, along with a few neighboring lakes, including Myrtle Lake, which serves as the primary outlet for the Grace Lake drainage system. The key calibration parameters encompassed horizontal and vertical hydraulic conductivity, and soil porosity. Initial soil data relied on the NRCS soil data bank, which is accessible through the NRCS website (<https://websoilsurvey.sc.egov.usda.gov/App/WebSoilSurvey.aspx>). This dataset comprised essential soil properties such as conductivity, soil porosity, saturated and field moisture content, and was available for various horizontal soil zones.

The observed gage data from the Seminole County WaterAtlas website (<https://seminole.wateratlas.usf.edu/>) were used to set the water levels at the beginning of the simulation for the main lakes, ponds, and wetlands, including Grace Lake and Myrtle Lake. Furthermore, the observed data from WaterAtlas was applied to evaluate the accuracy of the calibration and validation using several statistical metrics explained by Moriasi et al., 2007, 2015 and Davtalab et al., 2017; Coefficient of Determination (R^2), Nash-Sutcliffe Model Efficiency Coefficient (NSE), Mean Error (ME), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Maximum Flood Stage Difference (MFS D). Table 1 represents the metrics used by this study and the optimal value and range of each metric. Since these metrics are widely used, the formula is not presented here. This table shows the calibration and validation results for the Grace Lake water level as an example. Based on Table 1 and Fig. 5a, the calibration results for simulating Grace Lake water levels were satisfactory. Based on the graph and table, the simulation results for both the dry and wet periods were slightly lower than the observed data. The simulated maximum stage in Grace Lake, which was the maximum stage during Hurricane Irma, was 12.2 cm lower than the observed maximum stage. What is interesting about this is that the statistics improved for the validation simulation (Table 1 and Fig. 5b). For instance, the Nash statistic was 0.76 for calibration, which increased to 0.87 for validation. Also, the model was able to simulate a maximum stage for Grace Lake with less difference than the calibration step when compared to the observed maximum stage (6.7 cm difference).

Road flooding was observed around Myrtle Lake Hills Road on September 10, 2017, as recorded by the photo presented in Fig. 4. The final model simulated a flood stage of 15.5 M–NAVD–88 for just upstream of this road, which has a road crown elevation of around 15.2 M–NAVD–88. This means that the model accurately simulated road flooding for this location.

2.5. Climate change and Multi-Model ensemble

GCMs have been widely used to evaluate the impact of future climate change on regional hydrological models (Maurer & Hidalgo, 2008). In this study, seven GCM models under the moderate emission scenario

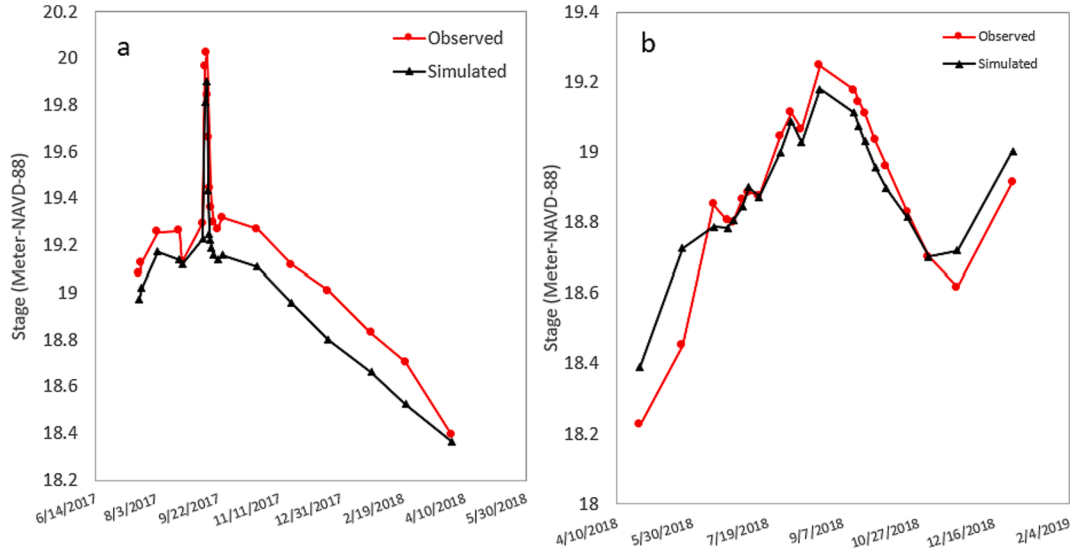


Fig. 5. Comparison of observed and simulated flood stage for Grace Lake; (a) calibration (b) validation.

RCP4.5 (Representative Concentration Pathway) were considered for projecting future climate data. These GCM data include BNU-ESM (Ji et al., 2014), GFDL-CM3 (Guo et al., 2014), MIROC-ESM (Watanabe et al., 2011), MPI-ESM-LR (Block and Mauritsen, 2013), MRI-CGCM3 (Yukimoto et al., 2012), NorESM1-M (Bentsen et al., 2013), and BCC-CSM1.1 (Xiao-Ge et al., 2013). The limitations of coarse resolution and biases in Global Climate Models (GCMs) hinder the generation of locally realistic climate variables essential for accurate input into hydrological models (Giorgi & Mearns, 2002; Wood et al., 2002). Hwang & Graham (2013) employed the Bias Correction and Stochastic Analog (BCSA) approach. This downscaling method provides daily data from 1976 to 2070 at a 12×12 km² grid within the study area. The data, obtained from the <http://ufdcimages.uflib.ufl.edu> website, were utilized in our study.

The differences in the structure of GCMs resulted in different projections of climate variables. The confidence of GCMs in projecting climate variables can vary regionally and spatially. In order to reduce errors in projected climate variables from GCMs, multi-model ensemble methods have been developed. Multi-model ensemble methods reduce the uncertainties in GCMs and enhance the reliability of the projected variables (Pavan & Doblas-Reyes, 2000; Giorgi & Mearns, 2002; Chandler, 2013; Wang et al., 2018). In this study, the probability analysis method with bounded distribution functions was used to deal with the uncertainties of the seven GCMs (Perkins et al., 2007; Gohari et al., 2013). The probability analysis method involved three main steps as follows:

Initially, the GCMs were weighted based on their abilities to estimate the observed climate variables from Sanford Gage (<https://www.weather.gov/wrh/climate?wfo=mlb>). To achieve this, the monthly precipitation values and temperatures were compared with the corresponding observed data recorded during the base period (1976 to 2005):

$$W_{ij} = \frac{\left(\frac{1}{\Delta d_{ij}}\right)}{\sum_{j=1}^N \left(\frac{1}{\Delta d_{ij}}\right)}$$

3. (1)

where W_{ij} is the weight of GCM j^{th} in month i in estimating monthly variable; Δd_{ij} is the difference between monthly observed climate variable and GCM j^{th} ; and N is the number of GCMs (i.e., seven here).

In the subsequent step, the computed weights of GCMs for the given month were used to develop the monthly discrete probability distribution function (PDF) of climate change variables, i.e., precipitation and temperature change. The monthly change in precipitation is defined as the ratio of mean monthly precipitation during the future period to that during the base period. The monthly change in temperature is defined as the differences between the mean monthly temperature during the future period and the base period. The monthly PDF of a climate change variable was then constructed to relate the mean monthly climate change variable to the weighting factor of that variable. This resulted in

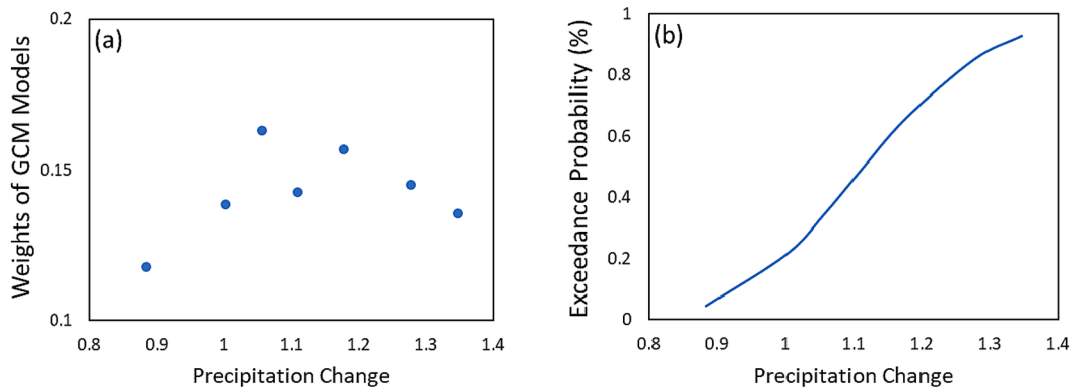


Fig. 6. The probability analysis of precipitation and generating: (a) discrete PDF of precipitation change, and (b) CDF of precipitation change in March.

Table 2
Average monthly and annual rainfall for base and future periods.

Month	Monthly and annual rainfall (mm)		
	Base (1976–2005)	Future (2040–2070)	Change
Jan.	80.0	90.5	13 %
Feb.	66.5	83.4	26 %
Mar.	110.6	114.1	3 %
Apr.	67.1	94.7	41 %
May	65.6	67.4	3 %
Jun.	193.9	225.6	16 %
Jul.	170.1	178.2	5 %
Aug.	188.8	214.1	13 %
Sep.	182.5	211.7	16 %
Oct.	102.5	113.0	10 %
Nov.	59.6	59.1	–1 %
Dec.	63.5	79.9	26 %
Annual	1351	1532	13 %
Wet month	838	943	13 %
Dry Month	513	589	15 %

a discrete PDF of climate change variables from the seven GCMs in a given month. For example, Fig. 6a shows the discrete PDF of precipitation in March.

The discrete PDFs for each month were then used to construct the continuous PDFs of climate change variables. Here, the Gamma distribution with two parameters, which have been successfully applied to describe the PDF of climate variables (Ines & Hansen, 2006; Piani et al., 2010; Msowoya et al., 2016), was used to model the continuous PDF of climate change variables from the discrete PDFs. The two parameters of climate change variables for each month were estimated by the maximum-likelihood method.

Finally, the developed monthly PDF of climate change variables was converted to the monthly cumulative distribution function (CDF). Fig. 6b shows the CDF of precipitation change in March. The lower (higher) exceedance probability of precipitation refers to the decrease (increase) of precipitation during the future period compared to the base

period, while the lower (higher) exceedance probability of temperature refers to the lower (higher) increase in temperature during the future period compared to the base period. Here, the 75 % probability percentile of precipitation and temperature change was used to characterize the precipitation and temperature during the base and future periods.

The weather generator (WG) is required to synthesize the daily climate variables from the monthly climate change variables obtained from probability analysis (Richardson, 1981; Richardson & Wright, 1984; Racsco et al., 1991). The Long Ashton Research Station Weather Generator (LARS-WG) has been successfully used to produce daily climate variables in diverse climates worldwide (Semenov & Barrow, 1997). In this study LARS-WG to generate the daily precipitation and minimum and maximum temperature for the 75 % probability percentile of monthly climate change variables.

The mean daily temperature (i.e., the mean of daily minimum and maximum temperature) was used to estimate the daily PET:

$$PET = 8 \times 10^{-6} \times T^{2.2684} \quad (2)$$

where PET is the daily potential evapotranspiration (inch/d), and T is the mean daily temperature (Fahrenheit).

Table 2 shows the projected monthly and annual rainfall for both the base (1976 to 2005) and future (2040 to 2070) periods. Based on this table the monthly rainfall has increased under the future conditions except in November, which has decreased slightly. Also, the amount of rainfall during the dry months (November to May) is expected to increase more than the increase in rainfall during the wet months (June to October).

4. Results and discussion

4.1. Climate change effects on surface water and groundwater interaction

Long-term continuous simulation offers valuable insights into the

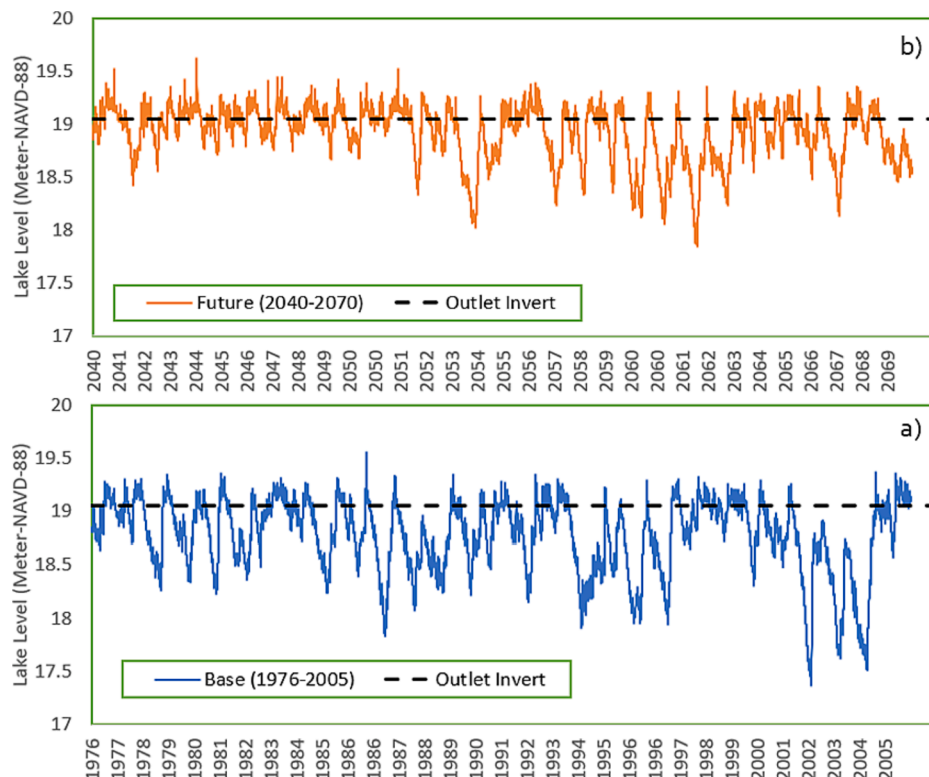


Fig. 7. Comparison of Grace Lake level for 30 years of continuous simulation of the base period (a) and future period (b).

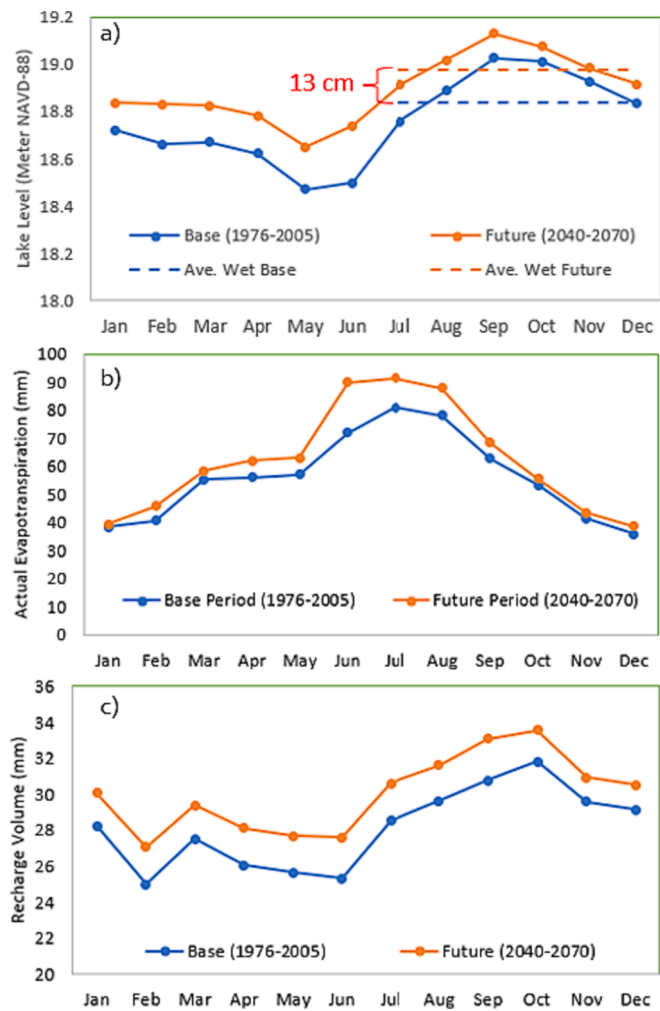


Fig. 8. Comparison of average monthly Grace Lake level (a), actual evapo-transpiration (b) and groundwater recharge (c) for base and future periods.

hydroperiods of study lakes and wetlands during the base period (1976–2005) and anticipated future changes (2040–2070). Fig. 7a and 7b show the comparison of 30 years of water level for Grace Lake for these two time periods. It’s worth mentioning that the water table aligns with the surface water elevation at Grace Lake. These figures depict that the duration of time that the lake level is below the outlet invert elevation (19.05 M–NAVD–88), or no spill duration, is significantly shorter for the future period. Given Grace Lake’s crucial role in flood mitigation, any spills that occur can be regarded as a failure of this stormwater pond. The reliability of stormwater ponds is defined as the number of days that the lake has spilled divided by the total days during the year (Davtalab et al., 2020). Using that definition, the average reliability of Grace Lake during the 30-year base model simulation was estimated at 74 %. The average reliability of this pond for the future condition was estimated at 61 %, which is a decline of approximately 13 % compared to the base period. The average reliability for Myrtle Lake has changed from 72 % for the base period to 54 % for the future period.

In fact, decreasing the reliability of a pond is a function of lake capacity for holding and retaining more flood. This capacity includes pond capacity and soil storage adjacent to the pond.

The long-term hydroperiod simulation conducted by ICPR model provides a comprehensive understanding of both current and future groundwater activities within the study area. The monthly average of the long-term surface water and groundwater simulation results are succinctly presented in Fig. 8a, 8b, and 8c. Fig. 8a shows the comparison of average monthly water levels for Grace Lake for the base period and

Table 3
Average groundwater recharge and actual evapotranspiration for base and future periods.

Month	Groundwater Recharge (mm)			Actual Evapotranspiration (mm)		
	Base	Future	Change	Base	Future	Change
Jan.	28.3	30.1	6 %	38.3	39.6	3 %
Feb.	25.0	27.1	8 %	40.7	46.0	13 %
Mar.	27.5	29.4	7 %	55.4	58.3	5 %
Apr.	26.1	28.1	8 %	56.1	62.0	11 %
May	25.6	27.7	8 %	57.1	63.0	10 %
Jun.	25.3	27.6	9 %	71.9	90.0	25 %
Jul.	28.5	30.6	7 %	81.1	91.6	13 %
Aug.	29.6	31.6	7 %	78.2	88.0	12 %
Sep.	30.7	33.1	8 %	62.7	68.4	9 %
Oct.	31.8	33.6	5 %	53.4	55.6	4 %
Nov.	29.6	30.9	5 %	41.5	43.7	5 %
Dec.	29.2	30.5	5 %	35.9	38.8	8 %
Annual	337.3	360.3	7 %	56.0	62.1	11 %

future period, based on the long-term simulation. Based on this figure, the lake’s water level is projected to increase during all months. While, in general, the water table plays an important role in the maintenance of the ecosystem (Russo & Lall, 2017), the average water level in wet months, or SHWL, directly has an impact on flooding. SHWL is normally used as the initial stage for synthetic storms or design storms which is explained in the next section. Based on this figure, SHWL for the future period is projected to increase by approximately 13 cm in Grace Lake.

Table 3 and Fig. 8a and 8b present the average monthly groundwater recharge and actual evapotranspiration during the base (1976–2005) and future (2070–2100) periods. The figure highlights a projected increase in recharge of approximately 7 % during both dry and wet periods in the future. Notably, the monthly recharge increase varies between 5 % and 9 %. These figures and the table illustrate that the rise in rainfall due to climate change leads to an increase in the water table within the soil column, consequently elevating the lake level. The higher water table, coupled with rising temperatures, intensifies evapotranspiration on the surface and in the unsaturated soil zone. This rise in the water table also has a feedback effect on the recharge amount. It is crucial to emphasize that the sinkhole plays a pivotal role, being the primary contributor to the majority of recharge within the study area.

Fig. 9 illustrates the depth to the water table at an example groundwater node, previously introduced in Fig. 3. To enhance the comprehension of groundwater changes from the base to the future period, the figure presents the exceedance probability for the depth to the water table. For instance, during the base period, the depth to the water table was approximately below 118 cm 50 % of the time, whereas this depth decreased to 97 cm during the future period. Alternatively, 100 % of the time during the base period, the depth to the water table was below 218 cm, and this depth was reduced to 187 cm during the future period. This graph effectively illustrates the decline in soil storage under future conditions, which can reduce the reliability of current stormwater management measures.

4.2. Evaluation of flood management measures

Stormwater structures are designed based on stationary observed data and they may not have the flexibility in a changing climate. Based on the economic value of projects or risk to human life, each stormwater structure should be designed to handle a specific storm or rainfall frequency (Panthou et al., 2014). To evaluate the flexibility of stormwater structure under climate change two 100-year 24-hour synthetic storms were simulated for the base and future periods. The calibrated ICPR model from previous sections were used to simulate the mentioned synthetic storms.

The rainfall intensity for each location within the USA is available from NOAA ATLAS 14 (NOAA, 2022). ATLAS 14 rainfall amounts are estimated based on the historical observed data. Also, Florida

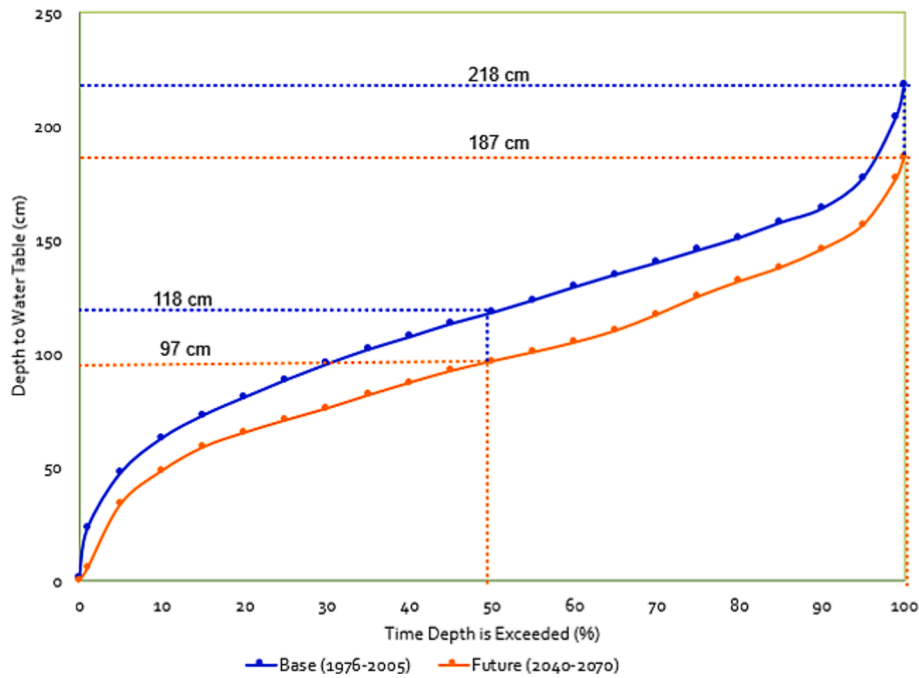


Fig. 9. Base period vs. Future Depth to Water Table Exceedance Probability.

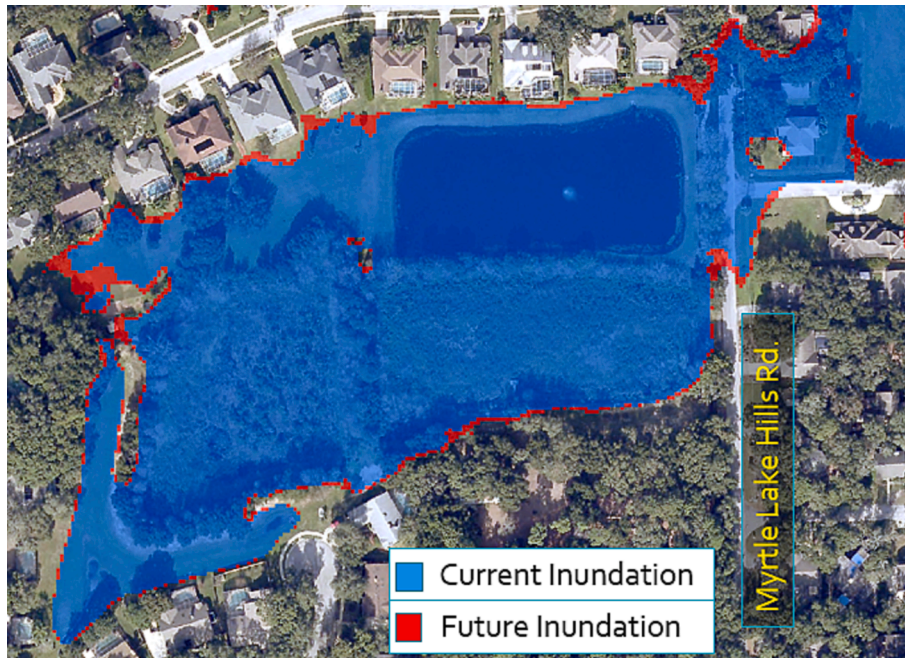


Fig. 10. Comparison of base and future inundated area (100 year-24 h storm).

International University (FIU) has recently conducted the future extreme rainfall projection for the entire state of Florida (Florida International University (FIU), 2021) using the same database used for Atlas 14. The future extreme rainfall amounts are estimated via a projecting change factor for Atlas 14 rainfall amounts for different time horizons. Based on the FIU 2021 study, the 100-year 24-hour storm for the study area was projected to have about a 20 % increase for the time horizon of 2060. The rainfall amount for the base period was 261 mm (100-year, 24-hour storm from NOAA-ATLAS 14 website) and future rainfall was 20 % higher than the base value, equating to 313 mm. For synthetic storms in domestic areas with significant storage, such as

Grace Lake, in addition to the rainfall amounts, the water level at the beginning of the design storm (initial stage) is important as well. Most of the time, SHWL (average water level during the wet months) provides a conservative estimation of initial water levels for design storms. For the base and future synthetic storms the initial stages are set to the average of the water levels for wet months during the base period (1976–2005) and future period (2040–2070), respectively.

To better understand the stormwater performance under base and future conditions, the flood stage for Myrtle Lake Hill Road was evaluated. This road had a flooding problem during Hurricane Irma, which dropped rainfall equal to an almost 100-year storm event within the

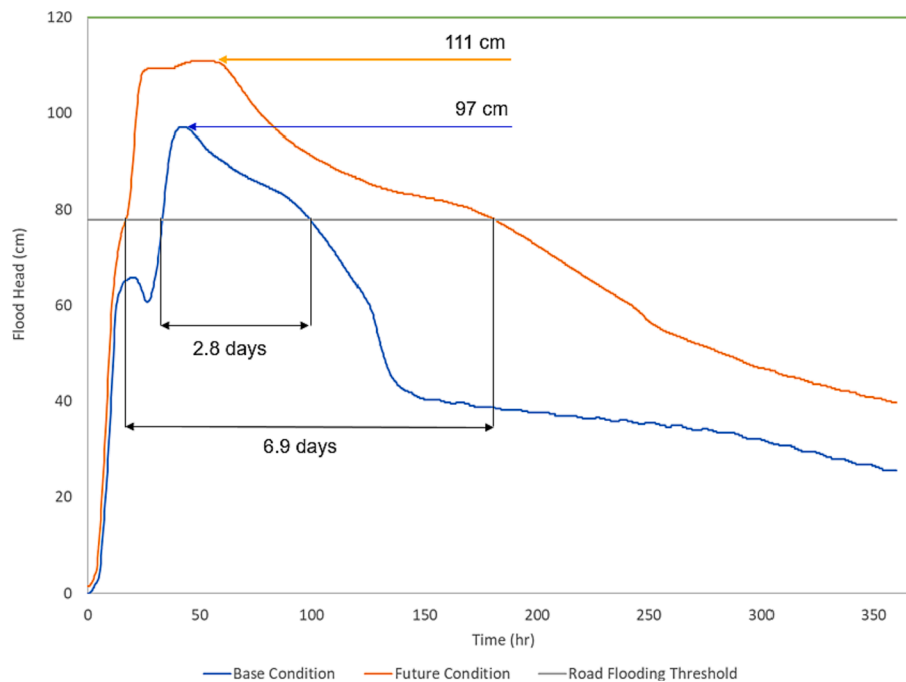


Fig. 11. Magnitude and duration of road flooding for Myrtle Lake Rd. (base and future 100-year 24-hour storm).

study area (254 mm). Fig. 10 shows the inundation area caused by the 100-year 24-hour flood for the base and future periods. Road flooding exists for both current and future conditions. The projected inundation area for the future period is slightly expanded compared with the base period, but the difference is minimal. Fig. 11 shows the base and future flood stage upstream of Myrtle Lake Road for the same storm event. Based on this figure, the depth of flooding on the road increased from 17 cm for the base period to 31 cm for the future period. Fig. 12 reveals that not only does flood stage increase under future conditions, but also the duration of road flooding increases. The 100-year, 24-hour storm caused two days and 18 h of road flooding for the base period, while future projection shows six days and 21 h of road flooding, which is more than 2.5 times longer. The intensified flooding issue on Myrtle Lake Hill Road results from the increased frequency of spills from Grace Lake in future conditions, coupled with the inadequacy of local storage to retain the additional runoff from both upstream and the local site.

Building on the extensive review conducted by this study, it is apparent that there are limited investigations into the impact of climate change on the reliability of stormwater ponds. However, a recent study by Davtalab et al. (2020) reported a notable decrease in the efficiency of retention ponds attributed to sea level rise and elevated groundwater. This finding aligns with the current results presented in this study.

4.3. Evaluation of water pollution load

Grace Lake and the connected upstream and downstream water bodies not only provide storage for flood mitigation, but also provide storage for treatment of pollutants. Florida has different criteria for estimating the treatment volume for any new development (FDEP, 2018). In preceding sections, it has been demonstrated that climate change exerts an impact on the future hydroperiod of Grace Lake, as well as on the capacity of the stormwater pond and adjacent soil storage. The decline in storage combined with higher rainfall intensity has led to the release of more pollutants from water bodies upstream, thus causing a water quality problem.

The Event Mean Concentrations (EMC) were used to estimate Total Nitrogen (TN) and Total Phosphorus (TP) loads at key locations (Perera et al., 2021). From the continuous simulation, the average annual

discharge for each water body was estimated and the average pollution concentration was used to estimate the total loads of TN and TP under current and future conditions (Annual pollutant = average concentration (mg/l) x annual discharge (mcm) x unit conversion).

Table 4 presents the - runoff volumes and nitrogen and phosphorus loads leaving Grace Lake during the 30-year continuous simulation during the base and future periods. On average, Grace Lake releases 203 thousand cubic meters of runoff annually, which contains approximately 147 kg/year of nitrogen and 6 kg/year of phosphorus, based on the simulation for the base period. The future projection for Grace Lake shows that it will discharge 339 thousand cubic meters per year of runoff, which will contain approximately 246 kg/year of nitrogen and 9 kg/year of phosphorus (about 67 % increases in nitrogen and phosphorus).

Fig. 12a and 12b show the annual average of the 30-year nitrogen and phosphorus discharges from Grace Lake to Myrtle Lake for the base and future periods. This schematic stormwater system displays the role of each stormwater pond or wetland in capturing upstream runoff, retaining some portion and releasing the rest downstream. Based on this figure, every year, Myrtle Lake receives approximately 299 kg of nitrogen and 36 kg of phosphorus on average for the current condition. The projected values for the future are approximately 389 kg of nitrogen and 48 kg of phosphorus per year (30 % increase in nitrogen and 33 % increase in phosphorus loads). Hence, the escalating flooding issues resulting from climate change can also contribute to water quality problems for the downstream wetland ecosystem. The findings on the environmental impacts of climate change and anthropogenic activities in this study align with several previous research efforts (Akbariye et al., 2019; Hou et al., 2021).

5. Conclusion

Florida experiences frequent tropical storms and is impacted by significant hurricanes every few years, such as Andrew (1992), Mathew (2016), Irma (2017), and Sally (2020). The state has implemented rigorous regulations for new developments to address flood mitigation and environmental protection in response to these natural hazards. However, despite these strong policies, climate data for most water

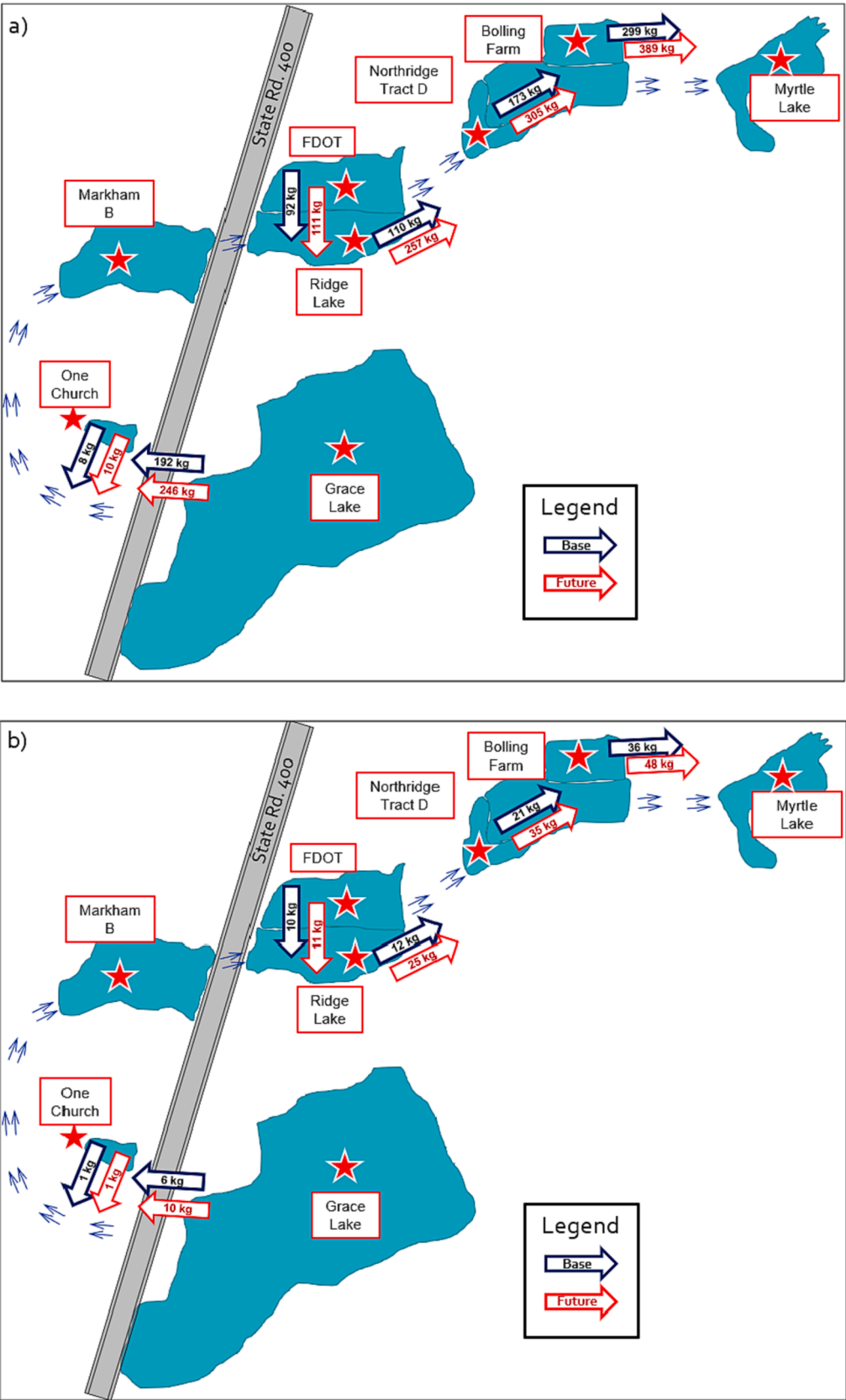


Fig. 12. Estimated average annual nitrogen (a) and phosphorus (b) discharge between Grace Lak and Myrtle Lake for the base and future periods.

Table 4
30-year summation and average annual of runoff, nitrogen, and phosphorus load releases from Grace Lake (comparisons of the base and future periods).

	Discharge Volume (1000 m ³)		Total Nitrogen (Kg)		Total Phosphorus (Kg)	
	Base	Future	Base	Future	Base	Future
30-year Summation	6085	10,163	4411	7368	170	285
Average Annual	203	339	147	246	6	9

districts within Florida are still assumed to be stationary (FDEP, 2018). Numerous studies, including the current research on Grace Lake, do not anticipate significant changes in future rainfall for Florida. For example, the current study projects only a modest 13 % increase. While this rise in rainfall may appear inconsequential, the long-term simulation of the Integrated Surface-Groundwater Model (ISGM) reveals a critical consequence: a 13 cm elevation increase in the seasonal high water. This heightened seasonal high water level becomes particularly critical during tropical storms, as it reduces soil storage capacity and diminishes the ability of stormwater ponds to capture stormwater runoff effectively. The study’s results demonstrate that the combination of increased future rainfall and higher seasonal high water elevations will lead to decreased reliability of Grace Lake as a stormwater pond for flood control. Currently, Grace Lake captures approximately 74 % of the annual storms on average, with projected decline to 61 % by mid-century.

Consequently, climate change may exacerbate the magnitude and duration of road flooding downstream of Grace Lake due to higher rainfall amounts and reduced soil and pond storage in future conditions. The event simulation revealed that the design storm of a 100-year 24-hour, experienced during Hurricane Irma, caused approximately three hours of road flooding downstream of Grace Lake. Future projections indicate that the duration of road flooding will nearly double for mid-century conditions. This finding highlights the potential for increased flooding issues in the middle of the current century, posing significant challenges for stormwater management in the state. Moreover, pollution loads are also affected by these changes, indicating that stormwater ponds like Grace Lake will have reduced capacity for water treatment. For instance, Grace Lake is projected to discharge about 67 % more nitrogen and phosphorus during the middle of the current century. Consequently, not only the reliability of stormwater ponds for flood control is expected to decline, but there will also be a notable failure in the effectiveness of water treatment processes. Notably, this study stands out as one of the few that utilizes ISGM to assess climate change impacts on both flooding issues and water quality.

In conclusion, the study’s findings underscore the urgency of adapting stormwater management strategies to address the challenges posed by climate change in Florida. The state must reevaluate and update its stormwater management systems to ensure resilience and effectiveness in the face of increasing flood risks and water quality concerns. Policymakers and environmental agencies should consider these results to develop proactive and sustainable solutions, safeguarding communities and the environment in the Grace Lake study area and beyond.

In this study, we encountered several limitations that warrant attention in future research endeavors. Our investigation focused on Grace Lake, a location plagued by significant sinkhole issues. The activation and collapse of sinkholes led to substantial fluctuations in lake levels across various time periods. Consequently, the extensive period covered by observed data proved less informative for our study, leaving us with only a few years of data for model calibration and verification. To enhance the robustness of our findings, it is advisable to replicate our methodology in other contexts with longer spans of observed data to validate the outcomes effectively.

While our results shed light on the significant challenges posed by both flooding and water quality, it is crucial to acknowledge the absence of consideration for the combined impacts of climate change and land

use changes in our study. As highlighted by Mahdian et al. (2023), the inclusion of land use changes could potentially exacerbate flooding issues. Therefore, it is imperative to incorporate land use change into future research to provide a more comprehensive understanding of the dynamics at play, particularly regarding the reliability of stormwater ponds.

Furthermore, the findings from a study by Davtalab et al. (2020) indicate that sea level rise may lead to groundwater rise, thereby diminishing soil storage and the reliability of stormwater ponds in low-lying areas. Given this insight, it would be valuable to extend our study to such regions and assess the combined effects of climate change and sea level rise on the efficacy of stormwater management strategies. By doing so, we can better anticipate and mitigate the potential challenges posed by evolving environmental conditions in vulnerable areas.

CRedit authorship contribution statement

Rahman Davtalab: Writing – original draft, Software, Methodology.
Saba Ghotbi: Writing – original draft, Methodology, Investigation.
Raana Koushki: Writing – review & editing, Validation, Supervision.
Caroline L. Reynolds: Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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